

The Ultra-Fast 4H-SiC LGAD with Etched Termination and Field Plate

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Abstract—The design, fabrication, and demonstration of a novel Silicon Carbide Low Gain Avalanche Detector (4H-SiC LGAD), exhibiting an ultra-fast time response and excellent time resolution, are reported. The use of field plates is proposed to suppress the high electric field caused by the negative bevel-etched angle in 4H-SiC LGADs based on TCAD simulations. Experimental measurements confirm that the field plate significantly increases the breakdown voltage of the 4H-SiC LGADs. Gain and time resolution are measured by using the ultraviolet transient current technique (UV-TCT) showing that 4H-SiC LGADs possess excellent timing performance, with a time resolution better than 35 ps for the injected laser signal with single minimum ionizing particle (MIP) charges generation at room temperature. Additionally, the gain suppression effect in a 4H-SiC LGAD is observed for the first time.

Index Terms—Silicon Carbide, LGAD, Ultra-Fast, UV-TCT

I. INTRODUCTION

SiC has received increasing attention as a potential replacement for silicon (Si) in charged particle detectors used in collider experiments to detect MIPs [1], [2]. However, due to challenges of growing ultra-low doped epitaxy ($< 5 \times 10^{13} \text{ cm}^{-3}$) for SiC, fabricating SiC devices with a thicker drift layer and low depletion voltage that meet the stringent demands of MIPs detection remains difficult. Devices with an internal gain can amplify MIP signals, thus overcoming the limitations of a thin drift region. In addition, as future collider detectors require precise 4D tracking (time and position), SiC LGADs stand out due to the excellent theoretical timing performance and better operability at room temperature, making it a potential alternative to Si LGADs.

Based on the design work [3], the 4H-SiC LGADs with a 75 μm drift layer has been successfully fabricated [4],

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demonstrating that electron-hole pairs generated by alpha particle ionization can produce low-gain avalanche multiplication within a wide operating voltage range. Since 4H-SiC LGADs are designed to measure MIPs, a relatively thick drift layer is required to generate a sufficient number of primary electron-hole pairs. However, the 75 μm thick drift layer makes it difficult for 4H-SiC LGADs to adopt the positive bevel-etched angle design used in mesa avalanching photodiodes (APDs) [5]. This issue could potentially be addressed in the future by advances in high-quality, large-area deep etching techniques for SiC. As a result, the bevel-etched 4H-SiC LGADs currently have to adopt a negative bevel-etched angle, which increases the junction electric field on the surface of bevel [6], leading to a reduction in the breakdown voltage of the device. In this work (Sec. II), we employed a field plate to suppress the electric field at the negative bevel-etched angle, thereby significantly increasing the breakdown voltage.

Due to the combination of the high signal-to-noise ratio (S/N) advantage of LGADs and the inherently faster carrier saturation velocity of SiC compared to Si, 4H-SiC LGADs theoretically should exhibit excellent timing performance. In this work (Sec. IV), we measured the time resolution of 4H-SiC LGADs using UV-TCT and demonstrated their excellent timing resolution for the first time.

II. DEVICE INTRODUCTION

Previous research on this 4H-SiC LGAD and the first measurements are reported in [4], and details of the fabrication process are in [7]. As shown in Fig. 1(a), the structure of the SiC LGAD mainly consists of three epitaxial layers: $(P^{++}) - (N^+) - (N^-)$. The 0.3 μm P^{++} layer with $1 \times 10^{20} \text{ cm}^{-3}$ serves to make an ohmic contact with the probe metal, the 0.5 μm N^+ layer functions as the gain layer, and the 75 μm N^- layer with $\sim 2 \times 10^{14} \text{ cm}^{-3}$ acts as the drift layer. In this work, as compared to [4], the gain layer doping is increased from $3 \times 10^{17} \text{ cm}^{-3}$ to $4 \times 10^{17} \text{ cm}^{-3}$ to obtain a higher gain factor, a field plate is added to prevent early breakdown, and the metal coverage area on the top layer is reduced to allow the 375 nm laser used for UV-TCT to penetrate into the drift layer of the device.

Fig. 2(a) shows two 600 μm diameter 4H-SiC LGADs, with the device on the bottom featuring a metal field plate. The detailed structure of the field plate is shown in Fig. 1(a), where the top metal electrode extends outward to form a field plate that covers and extends past the entire bevel-etched

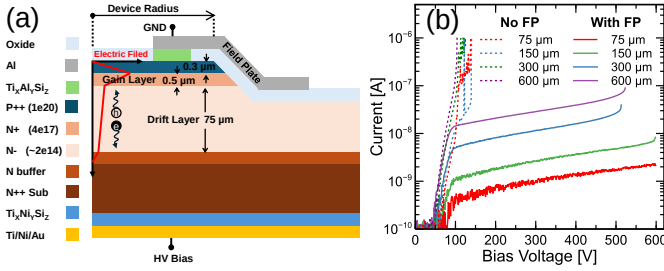


Fig. 1. (a) Schematic diagram of the 4H-SiC LGAD structure with field plate; (b) The comparison of measured I-V curves between 4H-SiC LGADs with and without a field plate.

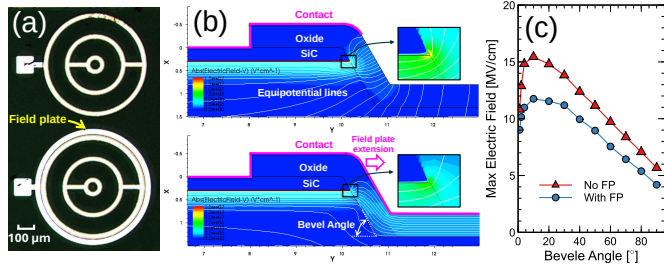


Fig. 2. (a) Two 600 μm diameter 4H-SiC LGAD without and with metal field plate; (b) Electric field distribution in the bevel-etched region obtained from TCAD simulation under a bias voltage of 500 V; (c) Maximum electric field on the surface of bevel-etched region as a function of the bevel angle.

surface. When the 4H-SiC LGAD operates, the applied bias voltage is connected to the bottom electrode, while the top electrode is grounded. Since the metal field plate acts as an equipotential body when grounded, the potential at the bevel-etched interface is reshaped, thereby suppressing the high electric field caused by the negative bevel-etched angle. Simulation results, as shown in Fig. 2(b), indicate that the electric field at the bevel-etched region with the field plate is lower compared to that without the field plate. Additionally, Fig. 2(c) demonstrates that the field plate effectively suppresses the electric field across bevel-etched angles ranging from 1 to 90 degrees.

Fig. 1(b) shows the measured reverse current-voltage (I-V) curves of 4H-SiC LGADs with and without field plates. To ensure consistency, the reversed I-V curves with diameters ranging from 75 μm to 600 μm are also included in Fig. 1(b). The results demonstrate a significant increase in the breakdown voltage for devices with field plates. This confirms that the field plate effectively suppresses the electric field, enhancing the breakdown voltage of the 4H-SiC LGADs. At a reverse bias of 600 V, for both the PIN diode and LGAD, the drift layer with a doping concentration of $\sim 2 \times 10^{14} \text{ cm}^{-3}$ and a thickness of 75 μm , is not fully depleted. To overcome this limitation and achieve higher breakdown voltages, future improvements could involve using epitaxial layers with lower doping concentrations or implementing better edge termination schemes, such as ion-implanted terminations or steeper mesa side walls.

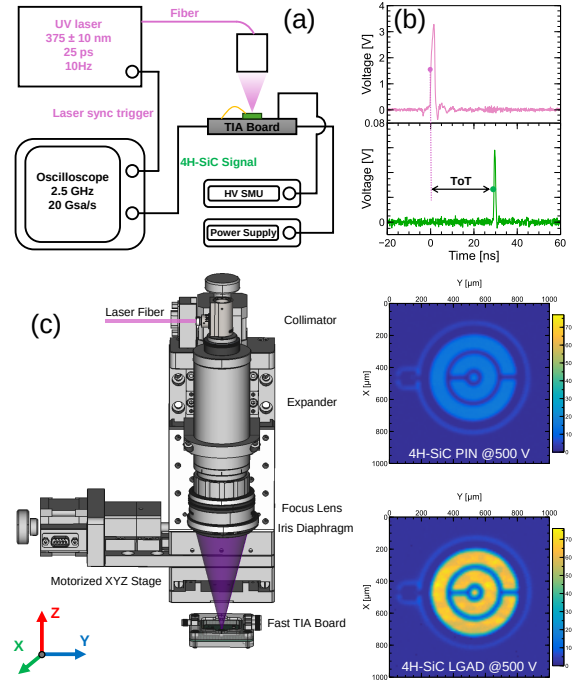


Fig. 3. (a) Schematic diagram of the UV-TCT system; (b) Oscilloscope-captured laser synchronization pulse and the signal induced by the laser in the 4H-SiC LGAD; (c) The construction of scanning and focusing system (left) and signal amplitude distribution of 4H-SiC PIN and 4H-SiC LGAD (right).

III. TEST SETUP

To investigate the gain and time resolution of the 4H-SiC LGAD, we constructed the UV-TCT system as shown in Fig. 3(a). The laser generates pulses with a wavelength of $375 \pm 10 \text{ nm}$ and a width of 25 ps, which are then output through a multimode optical fiber to a collimator and focused onto the 4H-SiC LGAD with $\sigma \sim 24 \mu\text{m}$ laser spot. The 4H-SiC LGAD is mounted on a readout board with a fast transimpedance amplifier (TIA). Finally, the synchronized trigger pulse from the laser and the signal pulse detected by the 4H-SiC LGAD are input into a digital oscilloscope with 2.5 GHz and 20 Gsa/s, shown in Fig. 3(b). In accordance with the method described in [4], we used a 4H-SiC PIN diode fabricated with the same process to study gain and timing performance. Fig. 3(c) shows the signal amplitude distribution of the 4H-SiC PIN diode and 4H-SiC LGAD, obtained by scanning with a focused laser spot driven by a motorized XYZ stage. In regions without metal coverage, the signal amplitude of the 4H-SiC LGAD is significantly higher than that of the 4H-SiC PIN diode.

IV. GAIN AND TIMING PERFORMANCE

The results of the charge collection shown in Fig. 4(a) indicate that, under the same laser injection intensity conditions, both the LGAD and the PIN diode show an increase in collected charge with increasing voltage; however, the collected charge of the LGAD is significantly greater than that of the PIN diode. The increase in voltage is accompanied by an increase in electric field, leading to an increase in gain factor.

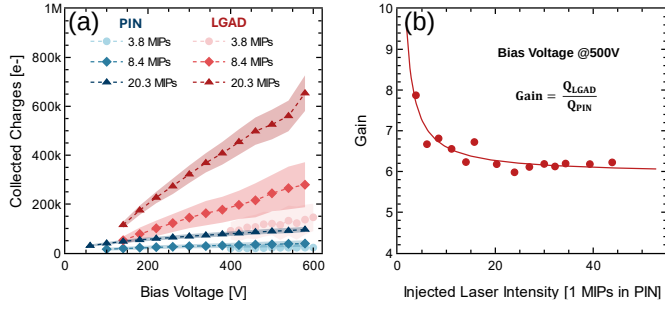


Fig. 4. (a) The change in the collected charge (the points are mean value and the color bands are widths of collected charges spectrum) of 4H-SiC PIN diode and 4H-SiC LGAD as a function of bias voltage; (b) The relationship between gain factor and laser injection intensity.

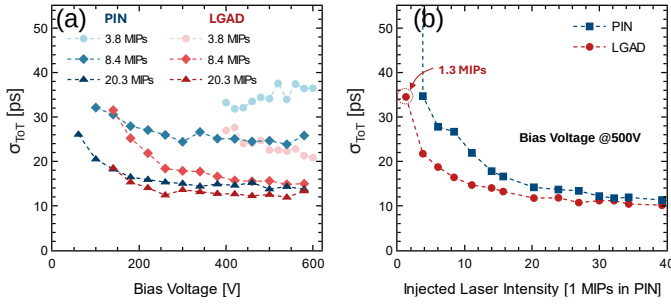


Fig. 5. (a) The variation of σ_{ToT} of 4H-SiC LGAD and 4H-SiC PIN as a function of bias voltage; (b) The relationship between σ_{ToT} and laser injection intensity.

Therefore, the LGAD exhibits electron-hole pair multiplication resulting in higher collected charge compared to the PIN diode. To best model the small signal scenario of MIPs, the laser intensity is calibrated by the collected charges from the 4H-SiC PIN diode to match the average charge generated by one MIP in $75 \mu m$ of 4H-SiC, which is $4275 e^-$ ($57 e^-/\mu m$) [8]. Fig. 4(b) shows the measured gain as the laser injection intensity increases. The gain suppression effect was observed in the 4H-SiC LGAD, which is similar to previous results from Si LGAD studies [9], [10], where the gain decreases due to the space charge effect by a large number of generated electron-hole pairs and high charge density.

To measure the time resolution, the synchronized laser trigger pulse was employed as the time reference with a small jitter contribution $\sigma_{trigger} = 7.7 ps$. As shown in Fig. 3(c), we collected 1000 pairs of pulses (the synchronized laser pulse and the 4H-SiC LGAD signal pulse) at each voltage level. The distribution of the time of transmission (ToT) was obtained using the constant fraction discrimination (CFD) method. A Gaussian function was then used to fit this distribution, ultimately yielding the σ of a Gaussian distribution which is the time resolution, denoted as σ_{ToT} . Fig. 5(a) shows the time resolution results of both the 4H-SiC LGAD and the 4H-SiC PIN diode with varying laser intensities. The time resolution of both devices improves as the bias voltage increases, but the LGAD exhibits a better time resolution compared to the PIN diode. Fig. 5(b) shows the measured time resolution as the laser injection intensity increases. These results demonstrate that the 4H-SiC LGAD possesses excellent

time resolution smaller than 35 ps, significantly outperforming the 4H-SiC PIN diode in measuring single MIP signals. As the laser intensity increases, the high S/N advantage of the 4H-SiC LGAD diminishes. At this point, the time resolution of both the LGAD and PIN becomes comparable, and is primarily limited by the sampling rate of the oscilloscope where $\sigma_{TDC} = \frac{50 ps}{\sqrt{12}} = 14.4 ps$ for 20 Gsa/s.

V. CONCLUSION

The fundamental electrical characteristics of the successfully fabricated 4H-SiC LGAD with etched termination and field plate were characterized. Using UV-TCT measurements, the gain and time resolution of the 4H-SiC LGAD were compared with those of the 4H-SiC PIN diode, demonstrating the excellent timing performance of the 4H-SiC LGAD for the first time, with a time resolution better than 35 ps for a single MIPs signal. The gain suppression effect in the 4H-SiC LGAD was observed for the first time. This work provides a promising approach for fabricating ultra-fast particle detectors for application in the future collider experiments.

REFERENCES

- [1] ECFA Detector R&D Roadmap Process Group, "The 2021 ECFA detector research and development road." doi: [10.17181/CERN.XDPL.W2EX](https://doi.org/10.17181/CERN.XDPL.W2EX).
- [2] B. Fleming and I. Shipsey, "Basic research needs for high energy physics detector research & development." doi: [10.2172/1659761](https://doi.org/10.2172/1659761).
- [3] P. Barletta, M. Cerullo, C. Haber, S. Holland, J. Muth, and B. Sekely, "Fast Timing With Silicon Carbide Low Gain Avalanche Detectors," 2022. arxiv: [2203.08554](https://arxiv.org/abs/2203.08554).
- [4] T. Yang, B. Sekely, Y. Satapathy, G. Allion, P. Barletta, C. Haber, S. Holland, J. F. Muth, S. Pavlidis, and S. Stucci, "Characterization of 4H-SiC Low Gain Avalanche Detectors (LGADs)," 2024. arxiv: [2408.12744](https://arxiv.org/abs/2408.12744).
- [5] L. Su, D. Zhou, H. Lu, R. Zhang, and Y. Zheng, "Recent progress of SiC UV single photon counting avalanche photodiodes," *Journal of Semiconductors*, vol. 40, no. 12, p. 121802, 2019. doi: [10.3389/j.1088/1674-4926/40/12/121802](https://doi.org/10.3389/j.1088/1674-4926/40/12/121802).
- [6] B. J. Baliga, *Fundamentals of power semiconductor devices*. Springer Science & Business Media, 2019. doi: [10.1007/978-3-319-93988-9](https://doi.org/10.1007/978-3-319-93988-9).
- [7] B. Sekely, Y. Satapathy, T. Yang, G. Allion, P. Barletta, C. Haber, S. Holland, J. F. Muth, S. Pavlidis, and S. Stucci, "Progress Towards 4H-SiC Low Gain Avalanche Detectors (LGADs)," in *Book of Abstracts from the International Conference on Silicon Carbide and Related Materials (ICSCRM 2024)*, (NC), pp. 694–695, Scientifi.Net, Sept 2024. doi: [10.4028/b-f6NMEP](https://doi.org/10.4028/b-f6NMEP).
- [8] M. Christanell, M. Tomaschek, and T. Bergauer, "4H-silicon carbide as particle detector for high-intensity ion beams," *Journal of Instrumentation*, vol. 17, no. 01, p. C01060, 2022. doi: [10.1088/1748-0221/17/01/C01060](https://doi.org/10.1088/1748-0221/17/01/C01060).
- [9] E. Currás, M. Fernández, and M. Moll, "Gain reduction mechanism observed in low gain avalanche diodes," *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, vol. 1031, p. 166530, 2022. doi: [10.1016/j.nima.2022.166530](https://doi.org/10.1016/j.nima.2022.166530).
- [10] G. Giacomini, W. Chen, G. D'Amen, E. Rossi, and A. Tricoli, "Spectroscopic performance of Low-Gain Avalanche Diodes for different types of radiation," *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, vol. 1066, 2024. doi: [10.1016/j.nima.2024.169605](https://doi.org/10.1016/j.nima.2024.169605).